

PROBABILITY AND RANDOM VARIABLES-1

PROBABILITY AND RANDOM VARIABLES-1 ▶ 29 November - 5 December ▶ الامتحان القصفي

Let A and B be disjoint events and assume that $P_r(A) = P_r(B)$ and $P_r(A \cup B) = 0.8$. Then $P_r(A) =$

Select one:

- ☐ a. 0.1
- ☒ b. 0.4
- ☐ c. 0.2
- ☐ d. 0.3

ABILITY AND RANDOM VARIABLES-1

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For a random variable X , suppose that $E(X^2) = 3$, then $E(2X^2 - 1) =$

Select one:

☐ a. 3

☐ b. 7

☒ c. 5

☐ d. 1

5

answered

out of 1.00

question

For a discrete random variable X , which of the following can be a c.d.f

Select one:

☐ a. $F(X) = \begin{cases} 0 & , X < 0 \\ \frac{1}{3} & , 0 \leq X < 2 \\ 1.1 & , X \geq 2 \end{cases}$

☒ b. $F(X) = \begin{cases} 0.1 & , X < 0 \\ \frac{1}{5} & , 0 \leq X < 2 \\ 1 & , X \geq 2 \end{cases}$

☐ c. $F(X) = \begin{cases} 0 & , X < 0 \\ \frac{1}{3} & , 0 \leq X < 1 \\ 1 & , X \geq 1 \end{cases}$

☐ d. $F(X) = \begin{cases} 0.1 & , X < 0 \\ \frac{1}{6} & , 0 \leq X < 2 \\ 1.1 & , X \geq 2 \end{cases}$

Let X be a continuous random variable with pdf $f(X) = e^{-\lambda X}$ when $X > 0$ and zero otherwise. The constant λ equals

Select one:

☐ a 0.6

☒ b 1

☐ c 0.5

☐ d 0.3

BILITY AND RANDOM VARIABLES-1

ABILITY AND RANDOM VARIABLES-1 ▶ 29 November - 5 December ▶ الامتحان النصفى

Let X be a continuous random variable. If $P_r(a < X < b) = 0.1$, Then $P_r(a \leq X < b) =$

Select one:

☐ a. 0.2

☒ b. 0.1

☐ c. 0.3

☐ d. 0.4

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Which of the following is **always** true

Select one:

- ☒ a. The variance of any random variable cannot be negative
- ☐ b. The correlation of two random variables can equal 2
- ☐ c. The covariance of any two random variables cannot be negative
- ☐ d. The correlation of any two random variables cannot be negative

PROBABILITY AND RANDOM VARIABLES-1

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Let X be some random variable with $Var(X) = 1$, then $Var(-1 - X) =$

Select one:

☐ a. 9

☐ b. 16

☒ c. 1

☐ d. 4

ABILITY AND RANDOM VARIABLES-1

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For a continuous random variable X which of the following is **wrong** about the c.d.f $F(X)$

Select one:

- ☐ a. $\lim_{t \rightarrow \infty} F(X) = 1$
- ☐ b. $\lim_{t \rightarrow -\infty} F(X) = 0$
- ☐ c. $F(X)$ is increasing
- ☐ d. $F(X)$ is a step function

Randomly we draw 3 balls from a box that contains 3 red balls and 2 black ball. The probability of choosing 2 black balls is

Select one:

☐ a. $\frac{\binom{3}{3}\binom{2}{0}}{\binom{5}{3}}$

☐ b. $\frac{\binom{3}{3}\binom{2}{1}}{\binom{5}{3}}$

☒ c. $\frac{\binom{3}{1}\binom{2}{2}}{\binom{5}{3}}$

☐ d. $\frac{\binom{3}{3}\binom{2}{1}}{\binom{5}{3}}$

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A continuous random variable X has c.d.f $F(x)$. For some numbers a and b with $a < b$ if $P_r(a < X < b) + F(a) = 0.3$ Then $F(b) =$

Select one:

☐ a. 0.1

☒ b. 0.3

☐ c. 0.2

☐ d. 0.4

ABILITY AND RANDOM VARIABLES-1

ABILITY AND RANDOM VARIABLES-1 ▶ 29 November - 5 December ▶ الامتحان المنمقي

Two independent random variables X and Y with $E(X) = 0.1$ and $E(Y) = 0.4$. Then $E(XY)$

Select one:

☐ a. 0.02

☒ b. 0.04

☐ c. 0.05

☐ d. 0.03



Two events A and B with $P_r(\bar{B}) = 0.4$ and $P_r(A - B) = 0.1$. Then $P_r(A|B) =$

Select one:

- ☐ a. $\frac{1}{4}$
- ☐ b. 1
- ☐ c. $\frac{1}{3}$
- ☐ d. $\frac{1}{2}$

Question 13

Not yet answered

Marked out of 2.00

Flag question

Two continuous and independent random variables X and Y with pdfs

$$f_X(x) = 1 \quad \text{when } 0 < x < 1 \quad \text{and zero otherwise}$$

$$f_Y(y) = e^{-y} \quad \text{when } y > 0 \quad \text{and zero otherwise}$$

Then the joint pdf is $f(x, y) =$

Select one:

☐ a.

$$\frac{e^{-y}}{4}$$

☐ b.

$$\frac{e^{-y}}{2}$$

☒ c.

$$e^{-y}$$

☐ d.

$$\frac{e^{-y}}{3}$$

ABILITY AND RANDOM VARIABLES-1

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For random variables X and Y , we have $E(X) = E(Y) = 1$ and $Cov(X, Y) = -4$. Then $E(XY) =$

Select one:

☐ a. -4

☐ b. -2

☒ c. -3

☐ d. -1

For an event A , we have $P_r(\emptyset | A) + 2 P_r(\Omega | A) =$

Select one:

☒ a. 4

☐ b. 5

☐ c. 3

☐ d. 2

PROBABILITY AND RANDOM VARIABLES-1

PROBABILITY AND RANDOM VARIABLES-1 ► 29 November - 5 December ► الامتحان المنهجي

Let X and Y be some random variables with joint pdf $f(x, y) = \begin{cases} \frac{1}{2} & , 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0 & , \text{Otherwise} \end{cases}$. The marginal pdf with respect to Y is

Select one:

☐ a. $\frac{x}{2}$

☒ b. 1

☐ c. $\frac{1}{2}$

☐ d. $\frac{1}{x}$

ABILITY AND RANDOM VARIABLES-1

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Let X and Y be some random variables with joint pdf $f(x, y) = \begin{cases} \frac{1}{2} & , 0 \leq x \leq 2, 0 \leq y \leq 1 \\ 0 & , \text{Otherwise} \end{cases}$. The marginal pdf with respect to Y is $\square$

Select one:

☐ a. $\frac{x}{2}$

☒ b. 1

☐ c. $\frac{1}{2}$

☐ d. $\frac{1}{x}$

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ABILITY AND RANDOM VARIABLES-1

PROBABILITY AND RANDOM VARIABLES-1 ▶ 29 November - 5 December ▶ الامتحان التمهيني

For random variables X and Y assume that $Cov(X, Y) = 2 Corr(3X, 2 + Y)$. Then $Var(X)Var(Y) =$

Select one:

- ☒ a. 4
- ☐ b. 16
- ☐ c. 9
- ☐ d. 25

PROBABILITY AND RANDOM VARIABLES-1

PROBABILITY AND RANDOM VARIABLES-1 ▶ 29 November - 5 December ▶ الامتحان التصفي

Let X and Y be some random variables with range $R_{XY} = \{(0,0), (0,1), (1,0), (1,1)\}$. If $f(0,0) = f(0,1) = f(1,0) = f(1,1)$. Then $f(1,1) =$

Select one:

☐ a. $\frac{1}{7}$

☒ b. $\frac{1}{4}$

☐ c. $\frac{1}{6}$

☐ d. $\frac{1}{5}$

For independent random variables X and Y , we have $Cov(1 - X, 2 + \pi Y) =$

Select one:

☒ a. $-\pi$

☐ b. 1

☐ c. 0

☐ d. π

For two events A and B with $P_r(A - B) = 0.7$ and $P_r(A \cap B) = 0.1$; then $P_r(A) =$

Select one:

- ☐ a. 0.9
- ☐ b. 0.6
- ☒ c. 0.8
- ☐ d. 0.7



Let X and Y be random variables with joint P.D.F

$$(x, y) = \begin{cases} x & , 0 \leq x \leq \sqrt{2}, 0 \leq y \leq 1 \\ 0 & , \text{Otherwise} \end{cases}$$

The marginal P.D.F with respect to X is $f_X(x) = x$; if $0 \leq x \leq \sqrt{2}$.

Select one:

☒ True

☐ False

Two continuous and independent random variables X and Y .

The marginal PDF of the random variable X is given by
 $f_X(x) = 1$ when $0 < x < 1$ and zero otherwise.

The joint PDF is $f(x, y) = 3e^{-3y}$ when $0 < x < 1$ and $y > 0$ and zero otherwise

Then the marginal PDF of the random variable Y is given by
 $f_Y(y) = e^{-3y}$ when $y > 0$ and zero otherwise

Select one:

☒ True

☐ False

Let X and Y be some random variables with joint P.D.F.

$$f(x, y) = \begin{cases} \frac{1}{4} & , 2 \leq x \leq \lambda, 2 \leq y \leq \lambda \\ 0 & , \text{Otherwise} \end{cases}$$

where $\lambda > 2$ is a constant ; then λ equals

Select one:

- ☒ a. 3
- ☐ b. 5
- ☐ c. 2
- ☐ d. 4

For some independent events A and B with $P_r(A) = P_r(B) = 0.1$, then we have $P_r(A \cup B) = 0.2$

Select one:

☐ True

☒ False

A continuous random variable X has
C.D.F $F(x)$ with
 $P_r(1 < X < 9) = 0.1$ then
 $(F(9) - F(1))^2 =$

Select one:

- ☐ a. 0.04
- ☐ b. 0.1
- ☒ c. 0.01
- ☐ d. 0.2

For some random variables X and Y ,
assume that $E(X) = 1$, $E(Y) = 2$ and
 $E(XY - X - Y) = 1$; then X and Y
are dependent.

Select one:



True



False

Let X and Y be some random variables with P.D.F f and range $R_{XY} = \{(0, 0), (0, 1), (1, 0)\}$. If $f(0, 0) = 0$ and $f(0, 1) = 3f(1, 0)$ then $f(1, 0) =$

Select one:

☐ a. $\frac{1}{5}$

☐ b. $\frac{1}{3}$

☒ c. $\frac{1}{2}$

☐ d. $\frac{1}{4}$



For two events A and B with
 $P_r(A - B) = 0.8$ and $P_r(A \cap B) = 0.1$
; then $P_r(A) =$

Select one:

- ☐ a. 0.7
- ☐ b. 0.6
- ☐ c. 0.8
- ☒ d. 0.9

For events A and B , if

$$P_r(A \cup B) = 0.5 \quad \text{and} \quad P_r(B) = 0.4$$

then $P_r(\bar{A} | \bar{B}) =$

Select one:

☐ a. $\frac{5}{7}$

☐ b. $\frac{5}{8}$

☒ c. $\frac{5}{6}$

☐ d. $\frac{5}{9}$

Randomly we draw 1 ball from a box that contains 9 red balls and 1 black ball. The probability that the ball is red equals

$$\frac{\binom{9}{0}\binom{1}{1}}{\binom{10}{1}}$$

Select one:

☐ True

☒ False



Let X and Y be two random variables with joint P.D.F

$$f(x, y) = 2u(x)u(y) e^{-x} e^{-2y}$$

then $f_Y(y) = u(y)e^{-2y}$

Select one:

☐ True

☒ False

$$\int_{-\infty}^{\infty} \sin\left(\frac{x+1}{x^2+1}\right) \delta(x) dx = \sin(2)$$

Select one:

☐ True

☒ False

For random variables X and Y , we have $E(X) = a$, $E(Y) = -a$ for some positive constant a . If $E(XY) = 1$ and $Cov(X, Y) = 2$; then $a =$

Select one:

- ☐ a. 4
- ☐ b. 3
- ☐ c. 1
- ☐ d. 2

Let X and Y be some random variables with P.D.F f and range

$R_{XY} = \{(0, 0), (0, 1), (1, 0)\}$. If $f(0, 0) = 0$ and $f(0, 1) = 2f(1, 0)$; then $f(1, 0) =$

Select one:

☒ a. $\frac{1}{4}$

☐ b. $\frac{1}{2}$

☐ c. $\frac{1}{3}$

☐ d. $\frac{1}{5}$

For a continuous random variable X with PDF $f(x) = \lambda e^{-4x}$ for $x \in (0, \infty)$ then the value of the co equals

Select one:

- ☐ a. 2
- ☐ b. 3
- ☐ c. 5
- ☒ d. 4



Two continuous and independent random variables X and Y .

The marginal PDF of the random variable X is given by

$f_X(x) = 1$ when $0 < x < 1$
and zero otherwise.

The joint PDF is $f(x, y) = 3e^{-3y}$
when $0 < x < 1$ and $y > 0$ and zero
otherwise.

Then the marginal PDF of the random variable Y is given by

$f_Y(y) = e^{-3y}$ when $y > 0$
and zero otherwise

Select one:

☐ True

☒ False

For some discrete random variable, which of the following can be a C.D.F.

Select one:

☐ a. $F(X) = \begin{cases} 0 & , X < 0 \\ \frac{1}{2} & , 0 \leq X < 3 \\ 1.01 & , X \geq 3 \end{cases}$

☐ b. $F(X) = \begin{cases} 0.3 & , X < 0 \\ \frac{1}{2} & , 0 \leq X < 3 \\ 1 & , X \geq 3 \end{cases}$

☐ c. $F(X) = \begin{cases} 0 & , X < 0 \\ \frac{3}{2} & , 0 \leq X < 3 \\ 1 & , X \geq 3 \end{cases}$

☒ d. $F(X) = \begin{cases} 0 & , X < 0 \\ \frac{1}{2} & , 0 \leq X < 3 \\ 1 & , X \geq 3 \end{cases}$

For random variables X and Y , we have $E(X) = a$, $E(Y) = -a$ for some positive constant a . If $E(XY) = 1$ and $Cov(X, Y) = 17$ then $a =$

Select one:

☐ a. 3

☐ b. 1

☐ c. 2

☒ d. 4

A discrete random variable X with range $\{1, 2\}$ and PDF

$$f(x) = P_r(X = x) = \frac{1}{3}\delta(x - 1) + a\delta(x - 2)$$

Then the positive constant a equals

Select one:

☐ a. $\frac{1}{2}$

☐ b. $\frac{1}{3}$

☐ c. $\frac{1}{4}$

☒ d. $\frac{2}{3}$

Let X and Y be some random variables with joint P.D.F.

$$f(x, y) = \begin{cases} \frac{1}{4}, & 1 \leq x \leq \lambda, 1 \leq y \leq \lambda \\ 0, & \text{Otherwise} \end{cases}$$

where $\lambda > 1$ is a constant ; then λ equals

Select one:

☒ a. 3

☐ b. 5

☐ c. 4

☐ d. 2

For a continuous random variable X with PDF $f(x) = \lambda e^{-4x}$ for $x \in (0, \infty)$ then the value of the constant λ equals

Select one:

☐ a. 3

☐ b. 5

☒ c. 4

☐ d. 2

Let X be some random variable with $Var(X) = a$, if $Var(aX) = 64$ then the positive constant a equals

Select one:

☒ a. 4

☐ b. 6

☐ c. 2

☐ d. 8

Let X be some random variable with $Var(X) = a$, if $Var(aX) = 64$ then the positive constant a equals

Select one:

☐ a. 2

☐ b. 8

☒ c. 4

☐ d. 6

Random variables X and Y with $\sigma_X \sigma_Y = 1$. We have
 $Cov(2X, 2Y) = 2 Corr(X, Y)$

Select one:

☐ True

☒ False



Random variables X and Y with $\sigma_X \sigma_Y = 1$. We have

$$\text{Cov}(2X, 2Y) = 2 \text{Corr}(X, Y)$$

Select one:

☐ True

☒ False